

Reflective Journal

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During this lab, our team explored the bias -variance tradeoff and learned how the complexity of a machine model affects its ability to make accurate predictions. We compared polynomial regression models with degrees 1, 4, and 15 to understand the differences between underfitting, a good fit, and overfitting. The degree 1 model was too simple and could not capture the curved pattern in the data, while the degree 15 model was too complex and fit the random noise instead of the overall trend. The degree 4 model provided the best balance because it followed the data without becoming overly complicated. We also created learning curves to compare training and cross-validation scores, which helped us better understand high bias and high variance.

This lab also helped us to better understand how model complexity affects the ability of a machine learning model to generalize to new data. Before completing this lab, we understood the definitions of underfitting and overfitting, but seeing the polynomial regression models and learning curves made the concepts easier to visualize and understand. The results showed me why a model that is too simple cannot learn enough from the data and why a model that is too complex may memorize the training data instead of generalizing to new data. Overall, this lab gave us a better understanding of how machine learning models are evaluated and why finding the right balance between simplicity and complexity is important when building models that perform well on unseen data.